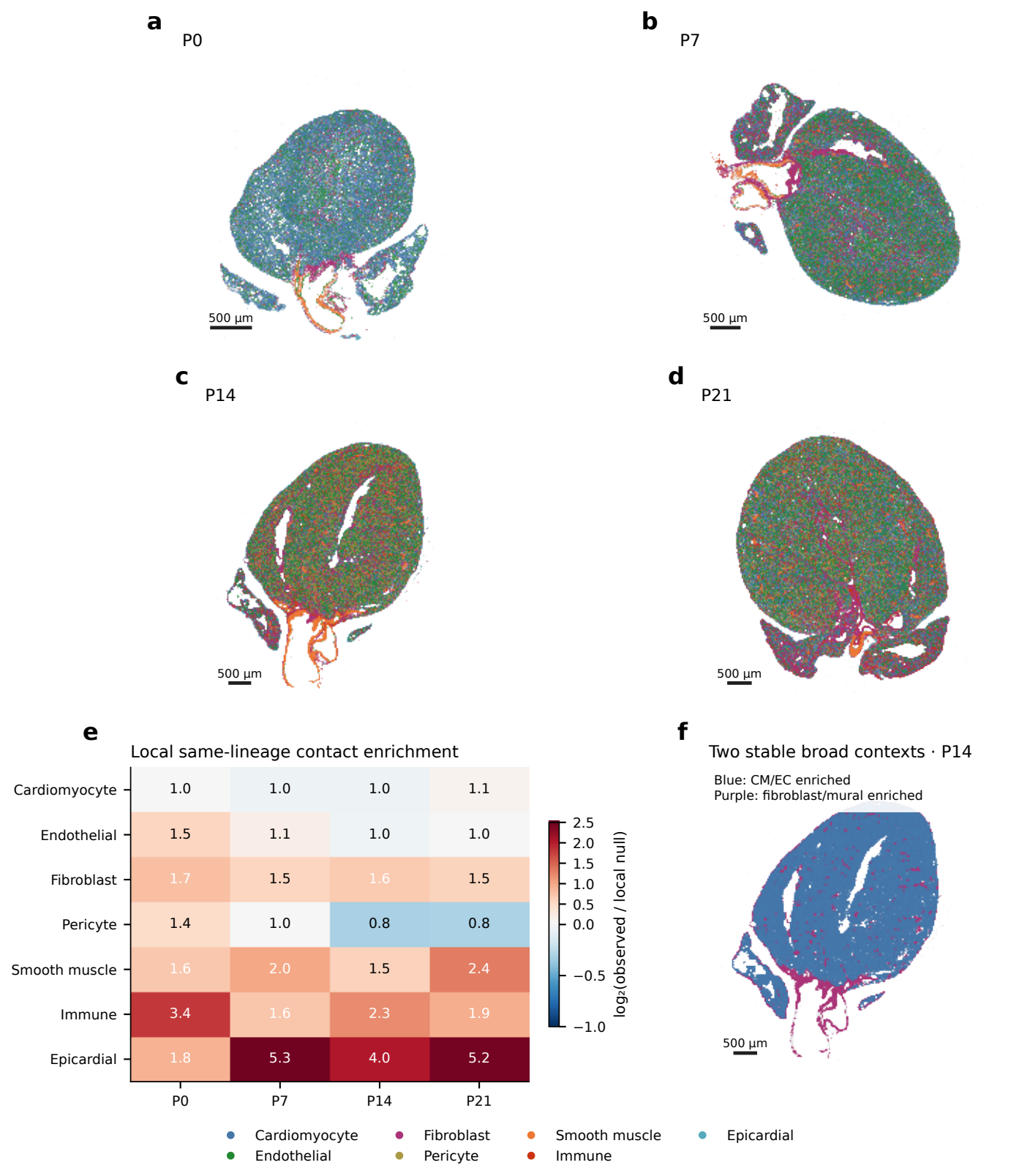
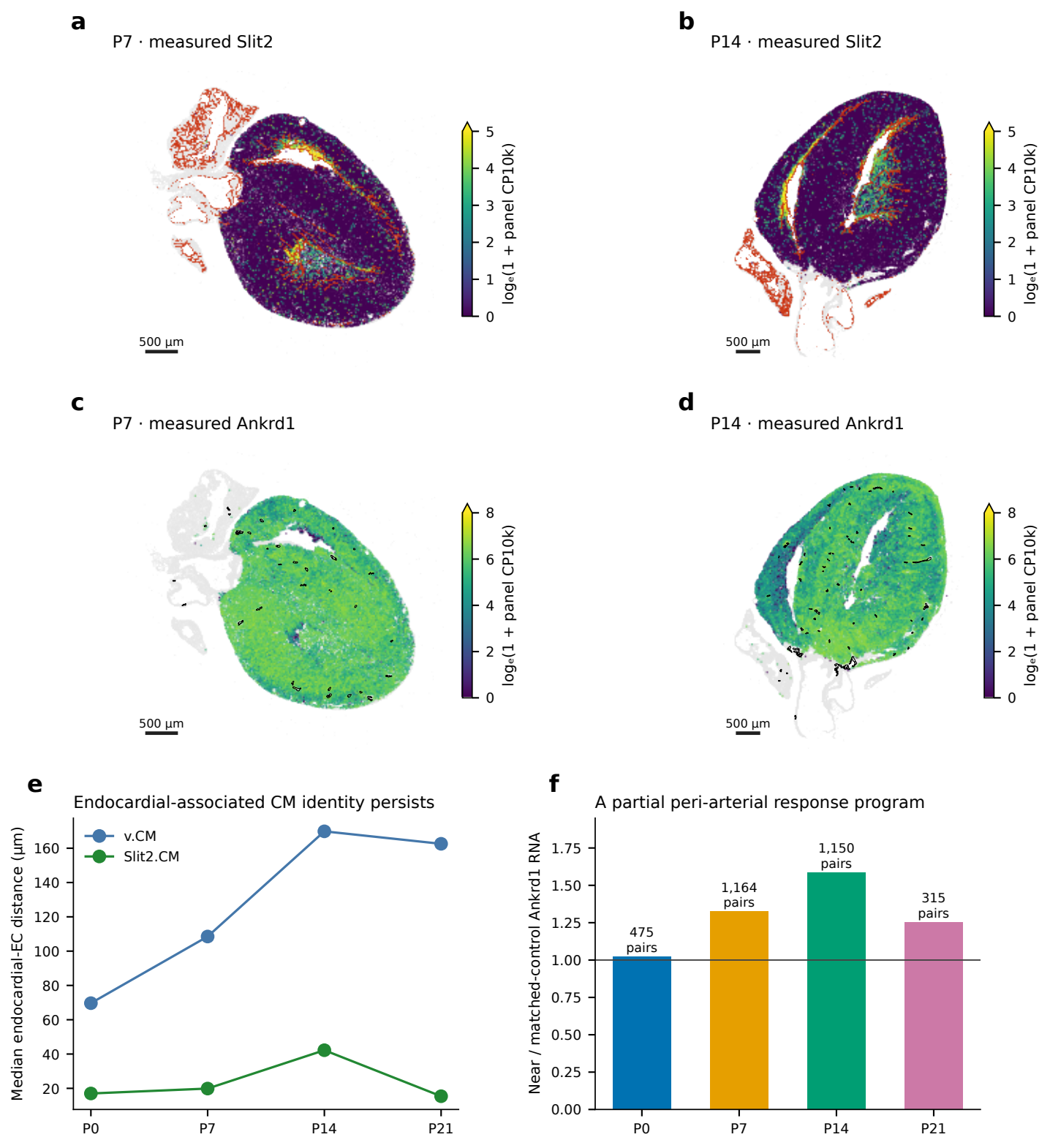
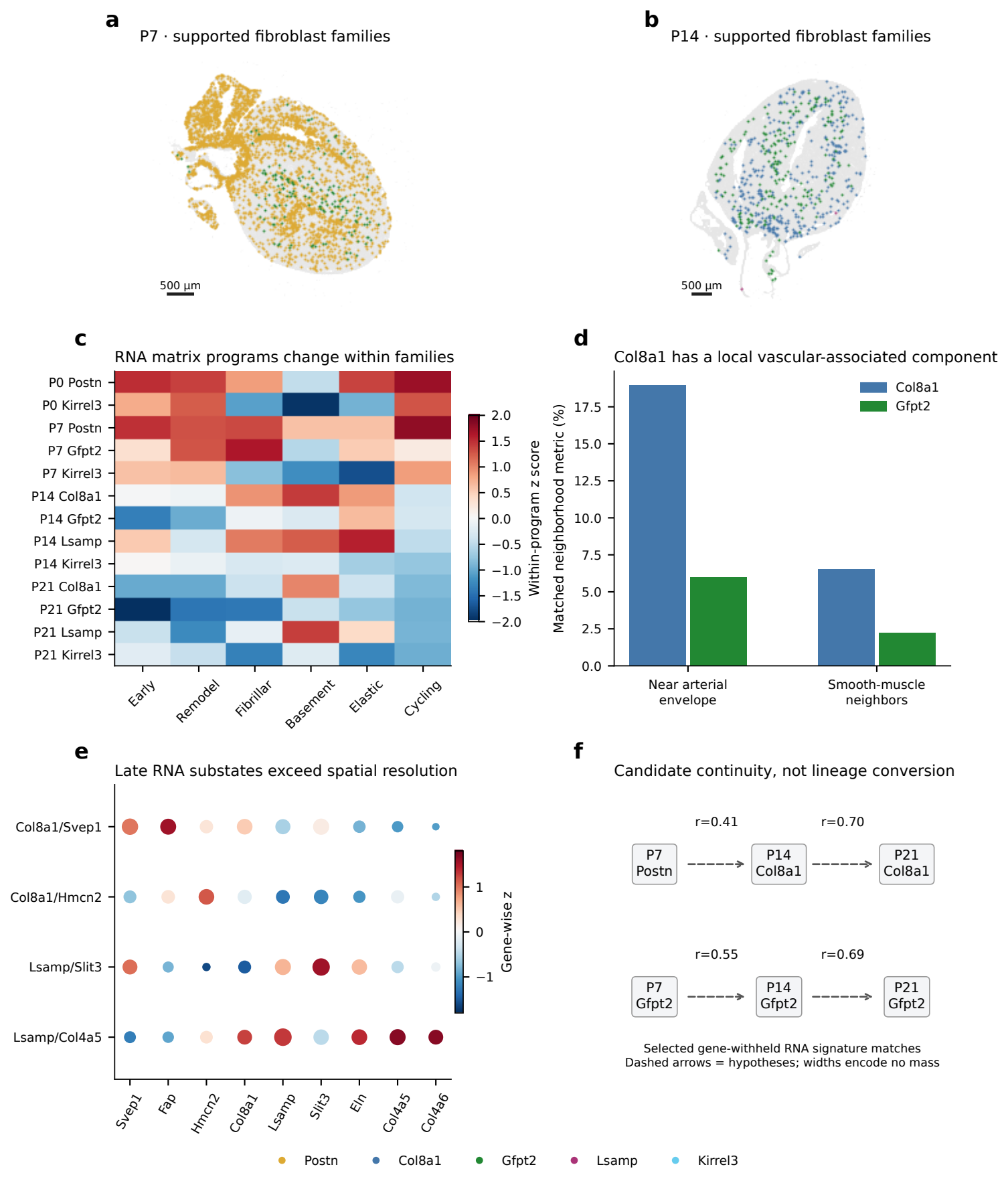
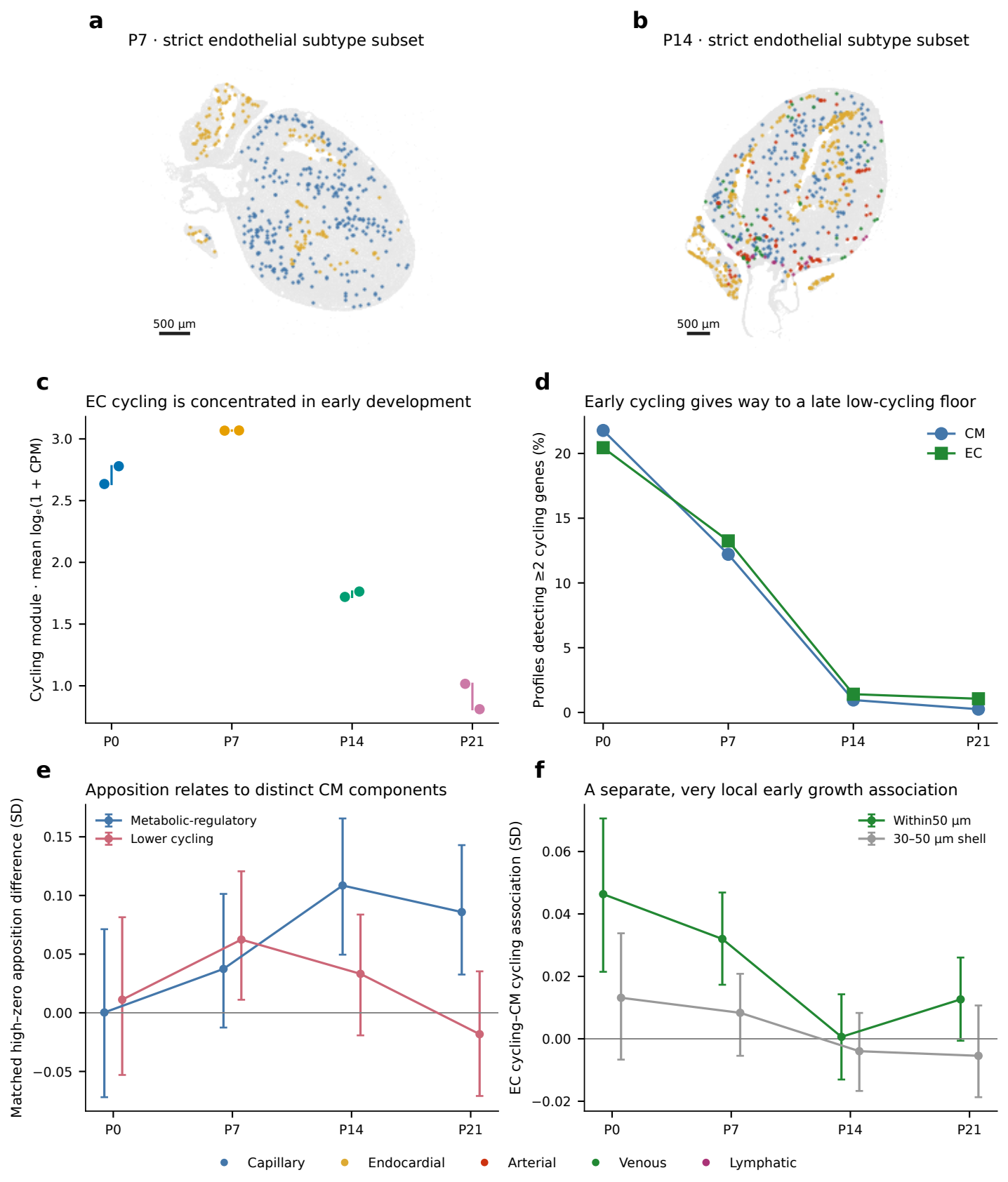
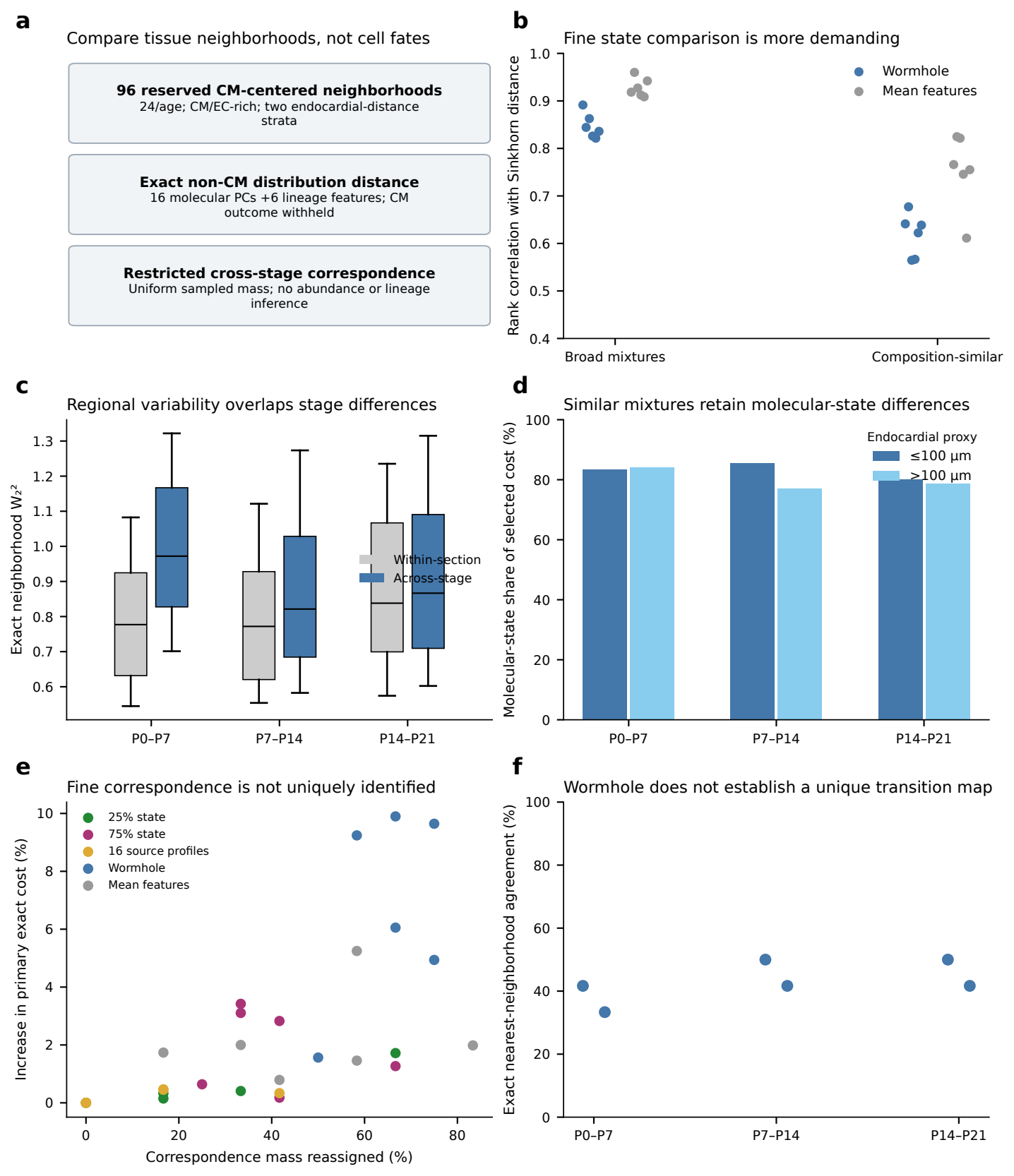

a
P0
500 µm
b
P7
500 µm
c
P14
500 µm
d
P21
500 µm
e
Local same-lineage contact enrichment
Cardiomyocyte
Endothelial
Fibroblast
Pericyte
Smooth muscle
Immune
Epicardial
1.0 1.0 1.0 1.1
1.5 1.1 1.0 1.0
1.7 1.5 1.6 1.5
1.4 1.0 0.8 0.8
1.6 2.0 1.5 2.4
3.4 1.6 2.3 1.9
1.8 5.3 4.0 5.2
P0
P7
P14
P21
log₂(observed / local null)
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
f
Two stable broad contexts · P14
Blue: CM/EC enriched
Purple: fibroblast/mural enriched
500 µm
Cardiomyocyte
Endothelial
Fibroblast
Pericyte
Smooth muscle
Immune
Epicardial


a
P7 · measured Slit2
500 µm
log$_e$(1 + panel CP10k)
b
P14 · measured Slit2
500 µm
log$_e$(1 + panel CP10k)
c
P7 · measured Ankrd1
500 µm
log$_e$(1 + panel CP10k)
d
P14 · measured Ankrd1
500 µm
log$_e$(1 + panel CP10k)
e
Endocardial-associated CM identity persists
v.CM
Slit2.CM
Median endocardial-EC distance (µm)
P0
P7
P14
P21
f
A partial peri-arterial response program
Near / matched-control Ankrd1 RNA
475 pairs
1,164 pairs
1,150 pairs
315 pairs
P0
P7
P14
P21

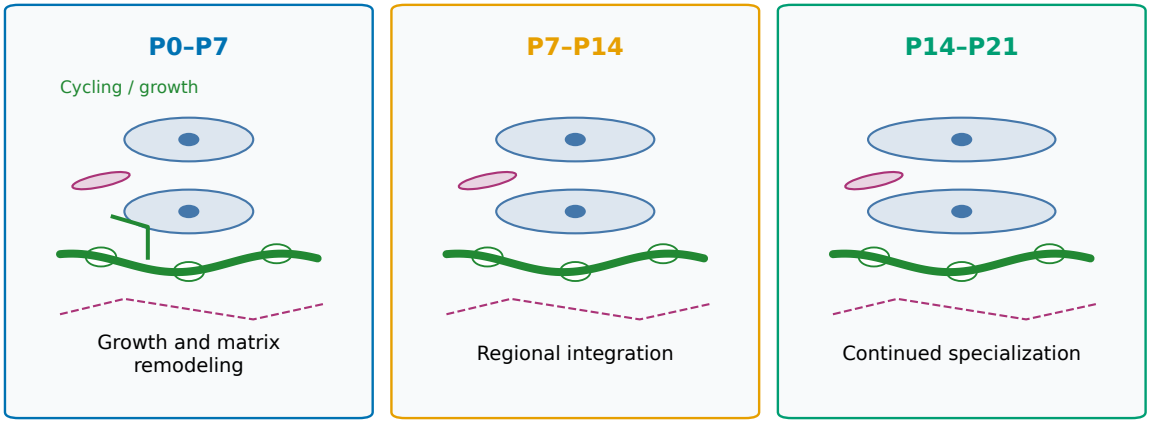

a
P7 · supported fibroblast families
500 µm
b
P14 · supported fibroblast families
500 µm
c
RNA matrix programs change within families
P0 Postn
P0 Kirrel3
P7 Postn
P7 Gfpt2
P7 Kirrel3
P14 Col8a1
P14 Gfpt2
P14 Lsamp
P14 Kirrel3
P21 Col8a1
P21 Gfpt2
P21 Lsamp
P21 Kirrel3
Early
Remodel
Fibrillar
Basement
Elastic
Cycling
Within-program z score
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
−1.5
−2.0
d
Col8a1 has a local vascular-associated component
Col8a1
Gfpt2
Matched neighborhood metric (%)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Near arterial envelope
Smooth-muscle neighbors
e
Late RNA substates exceed spatial resolution
Col8a1/Svep1
Col8a1/Hmcn2
Lsamp/Slit3
Lsamp/Col4a5
Svep1
Fap
Hmcn2
Col8a1
Lsamp
Slit3
Eln
Col4a5
Col4a6
Gene-wise z
1
0
−1
f
Candidate continuity, not lineage conversion
r=0.41
r=0.70
P7 Postn
P14 Col8a1
P21 Col8a1
r=0.55
r=0.69
P7 Gfpt2
P14 Gfpt2
P21 Gfpt2
Selected gene-withheld RNA signature matches
Dashed arrows = hypotheses; widths encode no mass
Postn
Col8a1
Gfpt2
Lsamp
Kirrel3


a
P7 · strict endothelial subtype subset
500 µm
b
P14 · strict endothelial subtype subset
500 µm
c
EC cycling is concentrated in early development
Cycling module · mean log$_e$(1 + CPM)
3.0
2.5
2.0
1.5
1.0
P0
P7
P14
P21
d
Early cycling gives way to a late low-cycling floor
Profiles detecting ≥2 cycling genes (%)
20
15
10
5
0
CM
EC
P0
P7
P14
P21
e
Apposition relates to distinct CM components
Matched high–zero apposition difference (SD)
0.15
0.10
0.05
0.00
−0.05
Metabolic-regulatory
Lower cycling
P0
P7
P14
P21
f
A separate, very local early growth association
EC cycling–CM cycling association (SD)
0.06
0.04
0.02
0.00
−0.02
Within50 µm
30–50 µm shell
P0
P7
P14
P21
Capillary
Endocardial
Arterial
Venous
Lymphatic


a
Compare tissue neighborhoods, not cell fates
96 reserved CM-centered neighborhoods
24/age; CM/EC-rich; two endocardial-distance strata
Exact non-CM distribution distance
16 molecular PCs +6 lineage features; CM outcome withheld
Restricted cross-stage correspondence
Uniform sampled mass; no abundance or lineage inference
b
Fine state comparison is more demanding
Rank correlation with Sinkhorn distance
Wormhole
Mean features
Broad mixtures
Composition-similar
c
Regional variability overlaps stage differences
Exact neighborhood W₂²
Within-section
Across-stage
P0–P7
P7–P14
P14–P21
d
Similar mixtures retain molecular-state differences
Molecular-state share of selected cost (%)
Endocardial proxy
≤100 µm
>100 µm
P0–P7
P7–P14
P14–P21
e
Fine correspondence is not uniquely identified
Increase in primary exact cost (%)
25% state
75% state
16 source profiles
Wormhole
Mean features
Correspondence mass reassigned (%)
f
Wormhole does not establish a unique transition map
Exact nearest-neighborhood agreement (%)
P0–P7
P7–P14
P14–P21

**a**

Working model · persistent places, changing states




Conceptual tissue units: blue CM; green EC/capillary; purple fibroblast/matrix. Shapes and stages are not morphometric measurements.

**b**

Anatomy conditions developmental responses

| Endocardial interface | Arterial / mural interface | Myocardial interstitium |
|---|---|---|
| Persistent Slit2-associated CM context; regional identity is not simply immaturity | Partial Ankrd1 response and Col8a1-associated stromal component | CM–capillary–pericyte neighborhoods with changing fibroblast and vascular states |

**c**

A discriminating experiment · oxygen × EC contact × CM state

| Delivery support | Direct contact | CM competence |
|---|---|---|
| Measure oxygen and respiration; equal EC number and matrix | Direct versus separated ECs; verify interface and viability | Early versus later starting states; calcium and contractile endpoints |

Independent donor preparations are inference units; perturb a candidate receptor only after a contact benefit is demonstrated